

Plant Biology

Plant Arbovirus Mobilizes a Vector Salivary Protein to Initiate Plant Infection

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Abstract

Plant arboviruses rely heavily on insects' feeding activities for successful transmission. Insect salivary proteins have been suggested to be essential for successful viral infection, but their exact mechanisms are largely unknown. In this study, we reveal that salivary factors from *Laodelphax striatellus* are necessary for infection of *Rice stripe virus* (RSV) in plants. A salivary carbonic anhydrase (LssaCA) is identified as an essential factor in promoting RSV infection. LssaCA interacts with a rice thaumatin-like protein (OsTLP) that has endo- β -1,3-glucanase activity and can degrade callose in plants. RSV infection induces callose deposition, which can be reversed by LssaCA. Furthermore, LssaCA directly binds to the RSV nucleocapsid protein (NP) in salivary glands, and the LssaCA-RSV NP complex still binds OsTLP and further increases its glucanase activity. This study provides new insights into the tripartite virus-insect vector-plant interaction, which is relevant to many agriculturally important plant arboviruses whose transmission is facilitated by insect salivary proteins.

eLife assessment

This study presents a **useful** set of experiments to test how a salivary protein might facilitate planthopper-transmitted rice stripe virus infection by interfering with callose deposition. If the conclusions can be confirmed, the study will enhance our understanding of the mechanisms mediating tripartite virus-insect vector-plant interactions and would be of general interest in plant science research. Whereas most experimental data is **compelling**, the conclusions rely on **inadequate** evidence for the salivary protein carbonic anhydrase being present two weeks post injection in the plant, which seems unlikely.

Introduction

During plant virus transmission by piercing-sucking insects, arboviruses are inoculated into the plant phloem via the secreted insect's saliva (Arca & Ribeiro, 2018; Conway et al., 2016; Wu, Yi, Lv, Mao, & Wei, 2022). Thus, insect saliva acts as an interface for the virus-insect-host

tripartite interaction and it can directly promote pathogen transmission to, and infection of, the hosts (Sun et al., 2020; Wu et al., 2022). However, despite the importance of insect salivary proteins in the tripartite interaction, there is still much to learn about how these proteins enable successful viral infection.

Previous studies have revealed that there are two ways in which insect saliva facilitates viral infection (Arca & Ribeiro, 2018). One is an indirect approach whereby the saliva modulates the host microenvironment at the feeding site, and saliva effectors work together to allow the arthropod to go unnoticed while it feeds on the host plant (Acevedo et al., 2019; Arca & Ribeiro, 2018; Sun et al., 2020). The other is direct regulation, whereby saliva proteins promote virus transmission through specific molecular interactions (Wen et al., 2019). Our recent work indicated that a *Laodelphax striatellus* mucin protein, LssaMP, enables the formation of the salivary sheath and facilitates the transmission of *rice stripe virus* (RSV) into the rice phloem (Huo et al., 2022). A recent study on leafhoppers revealed that the expression of a saliva calcium-binding protein is inhibited by *rice gall dwarf virus* (RGDV), thus causing an increase of cytosolic Ca^{2+} levels in rice and triggering callose deposition and H_2O_2 production. This increases the frequency of insect probing, thereby enhancing viral horizontal transmission into the rice phloem (Wu et al., 2022). Direct saliva protein-pathogen interactions have been reported in animal pathogens. For example, during transmission of *Borrelia burgdorferi* by *Ixodes scapularis*, the saliva protein Salp15 of *I. scapularis* binds to the bacterial outer surface protein C, which prevents the bacterium from being recognized by the animal immune system. In this way, the saliva protein enables the pathogen to infect the animal host (Ramamoorthi et al., 2005; Schuijt et al., 2008). The *Anopheles gambiae* saliva sporozoite-associated protein (AgSAP) binds to the surface of the *Plasmodium berghei* sporozoite. The binding of AgSAP to heparan sulfate inhibits local inflammatory responses, thereby facilitating early *Plasmodium* infection in the vertebrate host. Although most plant viruses are completely dependent on insect vectors for plant-to-plant transmission (Gray, 2008), the direct function of insect saliva proteins in mediating virus transmission remains largely uninvestigated.

During sap-feeding, arthropods produce two distinct types of saliva at different stages of the feeding process; gel saliva and watery saliva (Bonaventure, 2012; Lou et al., 2019). The former forms a salivary sheath to provide a smooth path for the stylet penetration (Lou et al., 2019). The latter is secreted into the phloem sieve elements to prevent them from plugging up, suppresses plant defense responses, induces pattern-triggered immunity (PTI) through herbivore-associated molecular patterns (HAMPs), and induces effector-triggered immunity (ETI) through avirulent gene, etc. (Huang, Liu, Xu, Bao, & Zhang, 2017; Ji et al., 2017; Yi, Wu, & Wei, 2021).

For phloem-feeding insects, callose deposited on phloem sieve tubes can function as a defense mechanism by reducing insect feeding and preventing long-distance viral movement (Hao et al., 2008; Hipper, Brault, Ziegler-Graff, & Revers, 2013; Will & Vilcinskis, 2015; Zavaliev, Ueki, Epel, & Citovsky, 2011). Callose is a β -(1,3)-D-glucan polysaccharide that is synthesized by callose synthases and degraded by β -(1,3)-glucanases. Plants can defend themselves by depositing callose at the sieve tubes in response to virus infection, whereas viruses can counter this by activating β -(1,3)-glucanases to degrade callose and unplug the sieve tube occlusions (Bucher et al., 2001; Hao et al., 2008; Wu et al., 2022; Zavaliev et al., 2011).

RSV is the causative agent of rice stripe disease, a serious disease of rice crops that has occurred repeatedly in China, Japan, and Korea (Xu, Fu, Tao, & Zhou, 2021). RSV is completely dependent on insect vectors for transmission to its host plants, and *L. striatellus* is the main vector ((Xu et al., 2021); Zhao et al., 2018). *L. striatellus* transmits RSV in a persistent-propagative manner. The virus initially infects the midgut, then disperses from the hemolymph into the salivary glands, and is inoculated into the plant host during *L.*

striatellus feeding (Huo et al., 2022). *L. striatellus* belongs to the order Hemiptera, whose members mainly feed from sieve tubes through their mouthparts (stylets) that penetrate plant tissues and reach sieve tubes to ingest the phloem sap (Tjallingii, 2006; van Bel & Will, 2016). RSV is secreted into the rice phloem via the watery saliva (Huo et al., 2022; Wang & Blanc, 2021).

In this study, we identified a molecular interaction among RSV, an *L. striatellus* saliva protein, and a plant β -1,3-glucanase. The insect saliva protein directly binds to the RSV nucleocapsid protein (NP) and then binds to a rice thaumatin-like protein to activate its β -1,3-glucanase activity. The activation of β -1,3-glucanase helps facilitate RSV infection of the host plant by inhibiting callose deposition in response to viral infection.

Results

L. striatellus salivary carbonic anhydrase is an essential factor in enhancing RSV infection

To investigate the influence of *L. striatellus* salivary proteins on RSV infection levels in rice plants, two experiments were conducted and compared: viruliferous insect feeding (VIF) and microinjection of purified virus (MPV) (Fig. 1A). The results showed that VIF caused high RSV infection levels and obvious rice stripe disease symptoms, while MPV did not lead to disease symptoms, and only a low level of RSV NP protein was detected inside the plant (Fig. 1B-E). These results suggested that the insect saliva factors are necessary for enabling RSV infection.

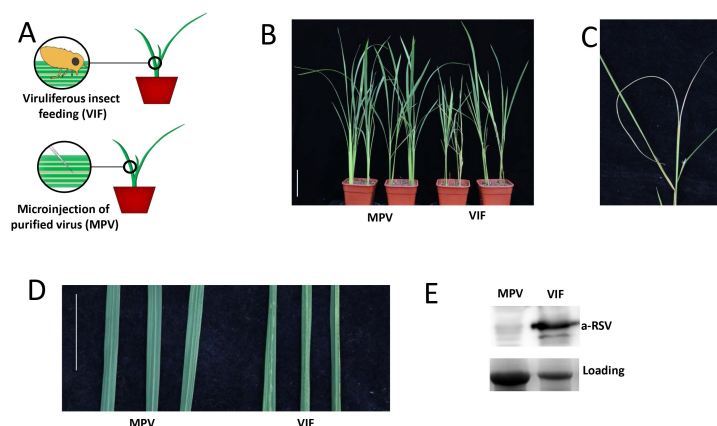


Figure 1.

Laodelphax striatellus feeding is necessary for *Rice stripe virus* (RSV) to infect rice.

A. The schematic diagram shows the two methods of rice inoculation with RSV: viruliferous insect feeding (VIF) and microinjection of purified virus (MPV). Three-leaf rice seedlings were used and viral infection was detected three weeks after inoculation.

B-D. Plant symptoms of rice stripe disease were observed in the VIF group (B-D), including heart leaf curling (C) and the striped spots on the leaves (D), but not in the MPV group (B and D). C shows an enlarged image of the VIF-group leaves.

A. E. Western blots to show the levels of RSV infection in plants inoculated with VIF or MPV. Anti-RSV antibodies were used to detect the RSV titers in the plants.

To identify the salivary proteins that play roles in mediating RSV infection, we have sequenced the *L. striatellus* saliva proteins and focused on the watery proteins that were inoculated into the plant phloem sieve tubes together with RSV (Huo et al., 2022). A salivary protein, Sap6 (GenBank accession No. RZF48846.1), was identified to have a high expression level in *L. striatellus* salivary glands (Fig. 2A). Western blotting assay showed that Sap6 was secreted into the plant as a watery saliva protein (Fig. 2B-C). Sequence homology analysis revealed Sap6 as a carbonic anhydrase-like protein, and enzyme active site analysis and enzyme activity experiment indicated Sap6 as a carbonic anhydrase (Fig. 2D-E). Sap6 was then named *L. striatellus* salivary carbonic anhydrase (LssaCA).

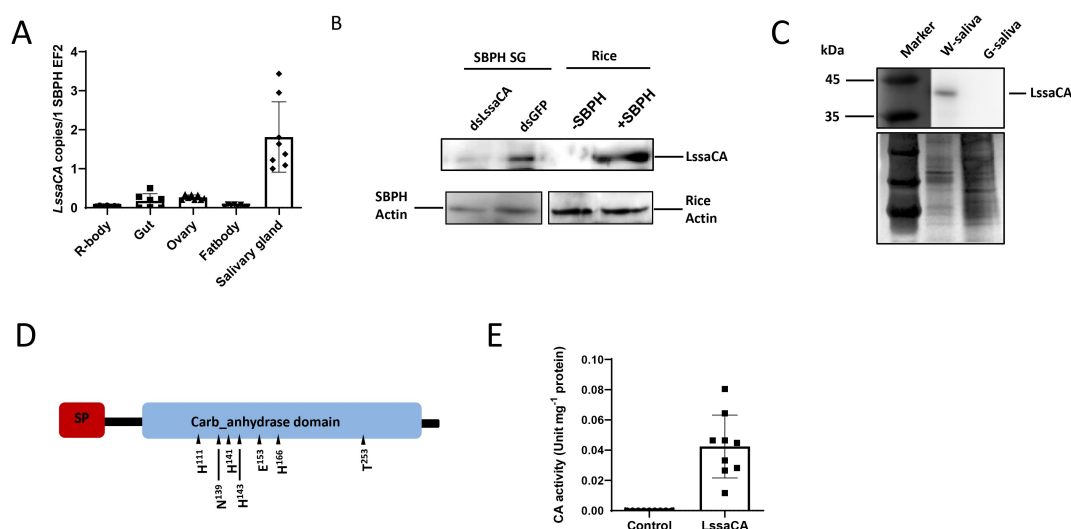


Figure 2.

Characteristics of the *L. striatellus* saliva carbonic anhydrase (LssaCA).

A. RT-qPCR analyses of *LssaCA* transcript levels in different tissues. R-body, the remaining body after removal of the guts, salivary glands, fat bodies, and ovaries.

B. Western blots showing LssaCA protein distribution in *L. striatellus* salivary glands (SBPH SG) and rice plants (Rice). The weaker signal in dsLssaCA-treated SBPH indicates a specific knock-down of LssaCA by dsLssaCA. Rice plants fed by SBPH acquire LssaCA (+SBPH), whereas control plants do not contain LssaCA protein (-SBPH).

C. Western blots showing the distribution of LssaCA in watery and gel saliva. Total watery or gel saliva proteins were detected by silver staining.

D. Schematic diagram of LssaCA protein sequence. SP, signal peptide; Carb_anhydrase domain, conserved eukaryotic-type carbonic anhydrase sequence. Triangles indicate seven predicted catalytically active residues.

E. Esterase activity of recombinantly expressed LssaCA protein.

To determine the role of *LssaCA* in RSV infection of rice plants, a *LssaCA*-specific double-stranded RNA (ds*LssaCA*) was synthesized and delivered into the hemocoel of RSV-infected 3rd instar nymphs to interfere with the gene expression. At 2 days post-microinjection (dpi), the *LssaCA* mRNA levels were measured by RT-qPCR and compared with the dsGFP control. The results showed that ds*LssaCA* treatment reduced the *LssaCA* mRNA level by 80% (Fig. 3A). The two groups of insects were then allowed to feed on healthy rice seedlings (5 insects per rice seedling) for 2 days. The RSV NP RNA levels (indicative of RSV titers) in the plants were measured at 14 days post-feeding (dpf). Compared with the rice seedlings fed by dsGFP-treated insects, those fed by ds*LssaCA*-treated insects had significantly reduced RSV titers (Fig. 3B), indicating that *LssaCA* played an essential role in mediating RSV infection of the rice plants.

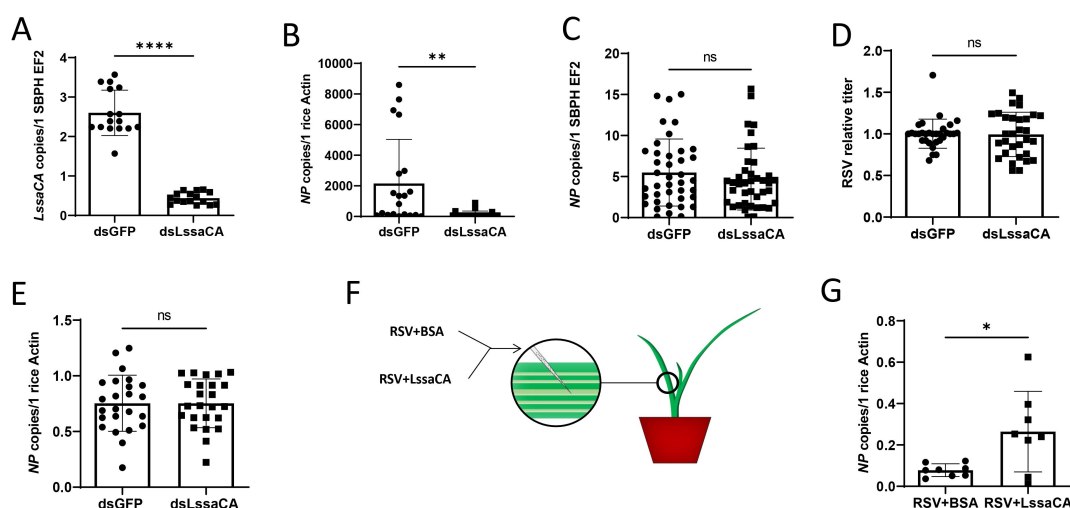


Figure 3.

The influence of *LssaCA* deficiency on RSV transmission and viral infection of plants.

A. The efficiency of gene silencing as determined by RT-qPCR. Each dot represents salivary glands from five RSV-infected third-instar nymphs.

B. RSV infection level in rice plants as determined by RT-qPCR. RSV infection levels are represented by the copy number of the *NP* gene. Each dot represents one rice plant fed on for 2 d and then grown for another 14 d.

C. RSV titers in *L. striatellus* salivary glands as determined by RT-qPCR. Each dot represents salivary glands from five RSV-infected third-instar nymphs.

D. RSV titers in *L. striatellus* saliva as determined by RT-qPCR. Each dot represents one saliva sample from 10 insects fed with an artificial diet.

E. RSV titers in rice plants as determined by RT-qPCR. Rice plants were fed for 24 h and RSV titers were assayed immediately post-feeding. Each dot represents one rice plant.

F. Schematic diagram to show the process of microinjecting an RSV particle solution into a plant. The RSV was pre-incubated with BSA or recombinantly expressed *LssaCA* protein before being microinjected into the plant phloem.

G. RT-qPCR to determine the level of RSV infection in microinjection-inoculated rice. The titers of RSV (copy number of *NP*) in rice plants were measured at 14 days post-inoculation (dpi).

BSA was used as a negative control. *, $p < 0.05$.

DsLssaCA and dsGFP indicate *LssaCA*- and *GFP*-specific dsRNA, respectively. ****, $p < 0.0001$; **, $p < 0.01$; ns, not significant.

LssaCA enhances RSV infection not before the viruses have been inoculated into the plant phloem

To determine at which step *LssaCA*-deficiency affected RSV infection, we analyzed the influence of *LssaCA*-deficiency on RSV titers during several steps of the RSV transmission process, including the virus load in the salivary glands and saliva, and the initial inoculation levels *in planta*. First, at 2 days after dsLssaCA treatment of the viruliferous insects when *LssaCA* was significantly downregulated, RT-qPCR was used to detect RSV titer in the insect salivary glands and the results showed that *LssaCA* deficiency did not reduce RSV titers in this tissue (Fig. 3C). Then, the *LssaCA*-deficient insects were allowed to feed on an artificial diet and their saliva was collected for RSV level analysis, which revealed that *LssaCA*-deficiency did not reduce the RSV titer in the secreted saliva (Fig. 3D). Finally, plants were fed with 10 viruliferous insects for 24 h respectively and the RSV titers were measured immediately after feeding, which showed that *LssaCA*-deficiency did not affect the RSV initial inoculation levels *in planta* (Fig. 3E). These results suggested that *LssaCA* promoted RSV infection by a mechanism occurred not in insects or early stage of viral entry in plants, but *in planta* after viral inoculation.

As shown in Fig. 1, VIF but not MPV is able to enable RSV infection of rice plants, suggesting the necessity of insect saliva factors in enabling RSV infection. To further confirm whether *LssaCA* played a role in directly enhancing RSV levels *in planta*, we produced a recombinant *LssaCA* protein to pre-incubate with purified RSV particle solution and used bovine serum albumin (BSA) as a control for comparison. The two groups were then microinjected into leaves on different plants, and at 14 dpi their viral titers were measured using NP gene-specific RT-qPCR. The results showed that compared to those inoculated with BSA plus RSVs particles, those inoculated with *LssaCA* plus these same particles exhibited significantly higher RSV titer values (Fig. 3F-G). These findings clearly indicate that *LssaCA* plays an important role in promoting RSV infection of plants after RSV has been inoculated into the plant.

LssaCA interacts with rice thaumatin-like protein to increase its endo- β -1,3-glucanase activity

Because *LssaCA* played its role *in planta*, we performed two experiments to investigate the mechanisms by which *LssaCA* enhances RSV infection. The first was to create *LssaCA* mutant proteins lacking enzymatic activity to determine if the enhanced infection was related to the *LssaCA* enzymatic activity. The second was to conduct a yeast two-hybrid screening to identify rice proteins that interact with *LssaCA*.

LssaCA has 7 residues constituting the enzymatic active center, including H₁₁₁, N₁₃₉, H₁₄₁, H₁₄₃, E₁₅₃, H₁₆₆, and T₂₅₃. Seven recombinant mutant proteins were constructed and were labeled as LssaCA^{H111D}, LssaCA^{N139H}, LssaCA^{H141D}, LssaCA^{H143D}, LssaCA^{E153H}, LssaCA^{H166D}, LssaCA^{T253E} (Fig. S2A). Enzymatic activity assays indicated that five of the seven mutants had significantly reduced enzymatic activity (Fig. S2B). To measure the influence of these mutants on RSV infection, they were pre-cultured with purified RSV particle solution and were then microinjected into plant leaves. The RSV titers were measured using RT-qPCR at

14 dpi. The results showed that, compared to LssaCA, none of the mutants decreased the protein function in enhancing RSV infection (Fig. S2C). These results suggest that LssaCA enables RSV infection in an enzymatic activity-independent manner.

The yeast two-hybrid screening identified a rice thaumatin-like protein (XP_025879846.1), designated as *Oryza sativa* thaumatin-like protein (OsTLP) (Fig. 4A-B and S3). The open reading frame of OsTLP encoded for a polypeptide of 327 amino acids and a thaumatin family (THM) domain from residues 27 to 267. A BLAST algorithm-based analysis against sequence databases revealed the sequence identities to TLP 1b. The N-terminal 24 amino acids constituted a secretion signal peptide, suggesting OsTLP as a secreted protein. Pull-down and MST assays were performed to confirm the molecular interaction between OsTLP and LssaCA. The pull-down assays showed that OsTLP and LssaCA had a specific interaction (Fig. 4C). The MST assay revealed a K_D of $1.3 \pm 2.2 \mu\text{M}$ for the OsTLP-LssaCA interaction, further confirming the molecular interaction (Fig. 4D).

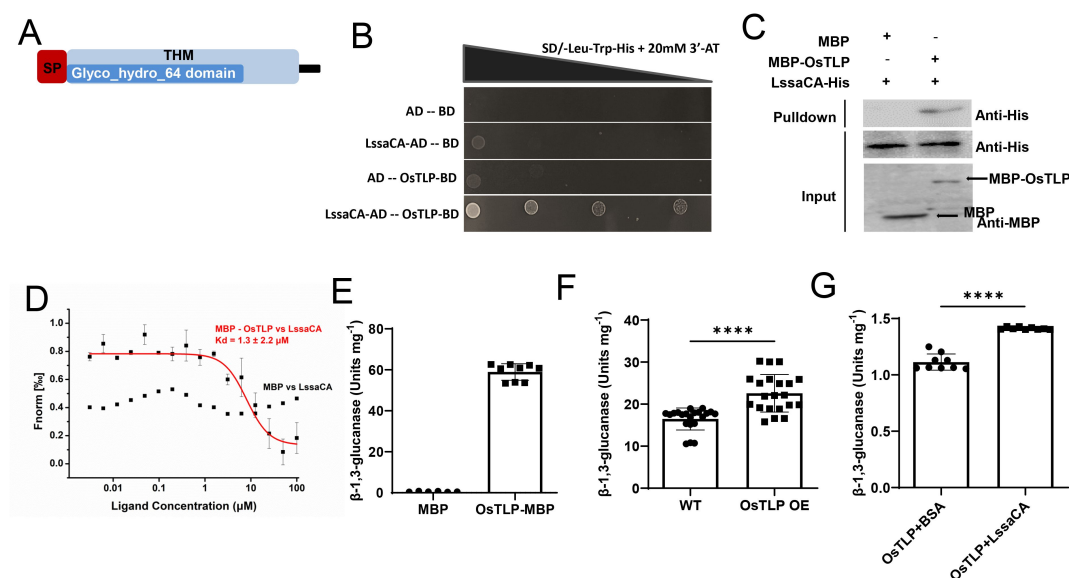


Figure 4.

The interaction between OsTLP and LssaCA plays a role in regulating the enzymatic activity of OsTLP.

A. Schematic diagram of the OsTLP protein sequence. SP, signal peptide; THM, conserved thaumatin family protein sequence. Glyco_hydro_64 domain, glycoside hydrolases of family 64.

B. Yeast two-hybrid assay showing the interaction between OsTLP and LssaCA. SD, synthetically defined medium; Leu, Leucine; Trp, Tryptophan; His, Histidine; 3'AT, 3-amino-1,2,4-triazole; AD, transcription activation domain; BD, DNA-binding domain.

C. Pull-down assays showing the interactions between LssaCA (LssaCA-His) and OsTLP (MBP-OsTLP). MBP was used as a negative control. Both anti-His and anti-MBP antibodies were used to detect corresponding proteins.

D. MST assay showing the interaction between LssaCA (LssaCA-His) and OsTLP (MBP-OsTLP). MBP was used as a negative control. Bars represent SE.

E. The endo- β -1,3-glucanase activity of the purified recombinant OsTLP protein (expressed as MBP-fusion protein). MBP was used to normalize results.

F. The endo- β -1,3-glucanase activity of OsTLP, which was overexpressed in transgenic plants (OsTLP OE) or wild-type plants (WT).

G. Regulation of OsTLP enzymatic activity by LssaCA. ****, $p < 0.0001$.

Previous studies have reported that TLP orthologs in apples, cherries, tomatoes, barley, etc. have endo- β -1,3-glucanase activity, which works on degrading callose (de Jesus-Pires et al., 2020). To confirm the endo- β -1,3-glucanase activity of OsTLP, an enzyme activity experiment was conducted with the recombinantly expressed OsTLP protein. The results revealed an average endo- β -1,3-glucanase activity of 60 units/mg OsTLP (Fig. 4E). To determine the *in-planta* activity of OsTLP, OsTLP overexpression transgenic plants were constructed. Enzymatic activity assays confirmed that the transgenic plants had significantly increased β -1,3-glucanase activity (Fig. 4F).

To investigate the influence of the molecular interaction between LssaCA and OsTLP on OsTLP activity, OsTLP was pre-incubated with either LssaCA or a BSA control for 2 h before the enzymatic activity was measured. The results showed that pre-incubation with LssaCA significantly increased OsTLP activity, compared to the BSA control (Fig. 4G). These results imply that LssaCA may facilitate RSV infection by enhancing the activity of OsTLP to degrade callose.

LssaCA inhibits callose deposition induced by RSV infection

Callose deposition on sieve plates is an important mechanism for slowing virus long-distance transmission and reducing the virus infection level in plants (Li et al., 2012; (Zavaliev et al., 2011)). To investigate whether RSV infection induces callose deposition, rice seedlings were fed by RSV-free or viruliferous *L. striatellus* for 24 h, and then callose deposition of the plants was measured. The results showed that VIF led to more callose deposition compared with that induced by feeding with RSV-free *L. striatellus* (Fig. 5A-B). ELISA assays also revealed an increased callose concentration in the plant fed by viruliferous *L. striatellus* (Fig. 5C). To further confirm the role of RSV in inducing callose deposition, the transcript levels of callose synthase genes were measured using RT-qPCR. We found that RSV infection upregulated *gsl3*, *gsl4*, *gsl5*, and *gsl10* (Fig. 5D-G). These results demonstrate that RSV infection induces callose deposition.

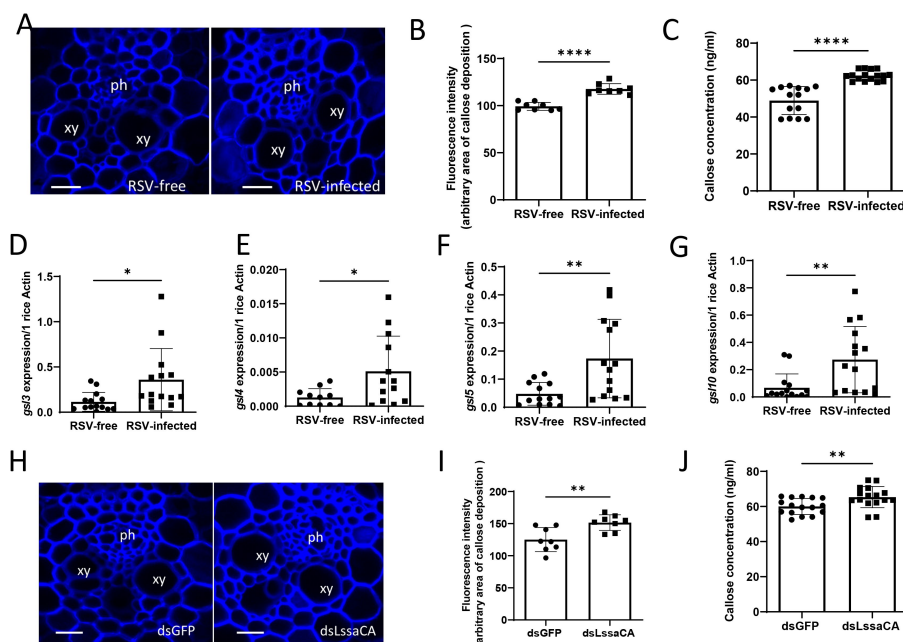


Figure 5.

LssCA inhibits RSV-induced callose deposition.

A. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples were prepared from the leaf phloem of plants fed on by RSV-free or RSV-infected *L. striatellus*. Thin sections were stained with 0.1% aniline blue at 24 h after *L. striatellus* feeding and examined under a fluorescence microscope. Xy, xylem; ph, phloem. Scale bars: 20 μ m.

B. Average fluorescence intensity of arbitrary area of callose deposition counted using ImageJ. Eight to ten random sites per sample were selected

ed for the evaluation of fluorescence intensity. ****, $p < 0.0001$.

C. Callose concentration in leaves of rice plants fed on by RSV-free or RSV-infected *L. striatellus*. ****, $p < 0.0001$.

D–G. Transcript levels of callose synthase genes as determined by RT-qPCR. Insects were allowed to feed on rice plants for 24 h before total RNAs were extracted. *, $p < 0.05$; **, $p < 0.01$.

A. H. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples were prepared from leaf phloem of plants fed on by dsGFP- or dsLssCA-treated RSV-infected *L. striatellus*. Thin sections were stained with 0.1% aniline blue at 24 h after *L. striatellus* feeding and examined under a fluorescence microscope. Xy, xylem; ph, phloem. Scale bars: 20 μ m.

I. Average fluorescence intensity of arbitrary area of callose deposition counted using ImageJ under a fluorescence microscope. Eight to ten random sites per sample were selected for the evaluation of fluorescence intensity. **, $p < 0.01$.

B. J. Callose concentration in leaves of rice plants fed on by dsGFP- or dsLssCA-treated *L. striatellus*. **, $p < 0.01$.

To determine if LssCA influences callose deposition induced by RSV infection, we downregulated *LssCA* in viruliferous *L. striatellus* by dsRNA microinjection. On 2 dpi when *LssCA* expression was significantly reduced, the insects were allowed to feed on healthy plants for 24 h, and then the callose deposition was measured. The results showed that the plants fed by *LssCA*-deficient insects deposited more callose compared to the plants fed by the control-group insects (dsGFP treatment) (Fig. 5H–J), indicating that *LssCA* can impair the callose caused by RSV infection.

OstLP degrades callose to facilitate RSV infection

To confirm if OstLP degrades the RSV-induced callose *in planta*, viruliferous *L. striatellus* were allowed to feed on either *OstLP* overexpression plants or control plants, and then callose deposition was measured. The results showed that transgenic plants exhibited significantly lower callose deposition compared to wild-type ones (Fig. 6A–B), indicating that

OstTLP can degrade the deposit callose induced by RSV infection. Moreover, we measured the RSV infection levels that were influenced by OstTLP overexpression. Viruliferous *L. striatellus* fed on either *OstTLP* overexpression plants or control plants for 2 days, and then the RSV titers were measured on 14 dpf. The RT-qPCR analyses indicated that, compared with wild-type plants, plants overexpressing *OstTLP* had significantly higher RSV titers (Fig. 6C), indicating that OstTLP facilitates RSV infection *in planta*. It is hypothesized that the overexpressed OstTLP degrades the deposited callose that is caused by RSV infection, therefore facilitating RSV infection.

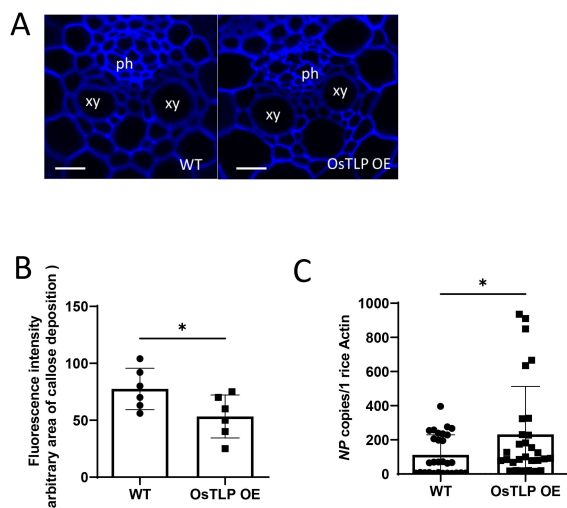


Figure 6.

OstTLP inhibits callose deposition to facilitate RSV infection.

A. Bright blue fluorescence of cross-sections showing callose deposition at feeding sites. Samples were prepared from the leaf phloem of plants fed on by RSV-infected *L. striatellus*. Scale bars: 20 μ m. OstTLP OE, transgenic plants over-expressing OstTLP.

B. Average fluorescence intensity of arbitrary area of callose deposition counted using ImageJ under a fluorescence microscope. *, $p < 0.05$.

C. Promotion of RSV infection by OstTLP overexpression, as determined by RT-qPCR. Titers of RSV (copy number of NP) were measured at 14 days post-feeding. *, $p < 0.05$.

RSV NP interacts with LssaCA to further enhance OstTLP enzymatic activity

We have confirmed that LssaCA binds to OstTLP to promote callose degradation and facilitates RSV infection *in planta*, we then investigated whether there is a direct interaction between LssaCA and RSV by which RSV directly participates in the function played by LssaCA. By using RSV and LssaCA-specific antibodies to conduct immunofluorescence assays (IFA), we found that LssaCA and RSV co-localized in salivary glands (Fig. 7A), suggesting a potential molecular interaction between them. A pull-down assay then revealed a specific interaction between LssaCA and RSV NP (Fig. 7B). An MST assay determined that the K_D of the interaction between LssaCA and NP was $2.7 \pm 2.2 \mu$ M, further confirming the molecular interaction (Fig. 7C).

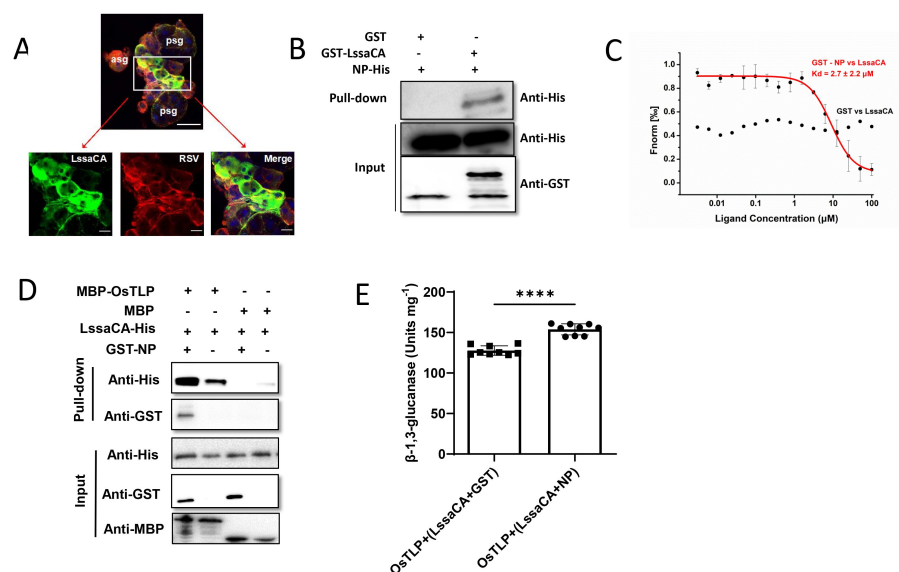


Figure 7.

LssaCA-RSV NP interaction enhances the OsTLP enzymatic activity.

A. Immunofluorescence assay showing co-localization of RSV (shown in red) and LssaCA (shown in green) in *L. striatellus* salivary glands. Images are representative of three independent experiments with a total of 30 SBPHs analyzed. The scale bar represents 20 μm . psg, principal salivary gland; asg, accessory salivary gland.

B. GST pull-down assays show the interaction between LssaCA (GST-LssaCA) and RSV NP (NP-His). GST was used as a negative control. Anti-

GST and anti-His antibodies were used to detect the corresponding proteins.

C. MST assay showing the interaction between LssaCA (LssaCA-His) and RSV NP (GST-NP). GST was used as a negative control. Bars represent SE.

D. Pull-down assays showing the interaction between OsTLP (MBP-OsTLP) and LssaCA-NP complex. LssaCA was expressed with a His tag and RSV NP was expressed as a GST-fusion protein. LssaCA and NP were preincubated before co-incubation with OsTLP. Anti-His and anti-GST antibodies were used to detect the corresponding proteins.

E. Regulation of OsTLP enzymatic activity by LssaCA and LssaCA-NP. (LssaCA + NP) indicates that the two proteins were pre-incubated before incubation with OsTLP. ****, $p < 0.0001$.

As LssaCA interacts with both RSV NP and OsTLP, we measured the tripartite interaction. Through pull-down assays, we discovered that when LssaCA was pre-incubated with RSV NP to form an NP-LssaCA complex, the complex still had the ability to bind to OsTLP (Fig 7D). This result demonstrates a tripartite interaction among these three proteins.

We then explored how the tripartite interaction affects OsTLP enzymatic activity. OsTLP was pre-incubated with LssaCA-NP complex or LssaCA control. After incubation for 2 h, the enzyme activity was measured. The results showed that pre-incubation with LssaCA-NP complex significantly increased OsTLP activity compared with LssaCA control (Fig. 7E), indicating that stronger activation of β -1,3-glucanase activity was achieved through the NP-LssaCA-OsTLP tripartite interaction.

Taken together, these results suggested that when RSV is delivered into plant phloem, it can induce callose deposition. This process is overturned through a tripartite interaction involving the LssaCA protein which binds to RSV in insect salivary glands and then the LssaCA-RSV complex binds to OsTLP in plants. Once this tripartite interaction occurs, OsTLP activity of β -1,3-glucanase is significantly increased which degrades callose thus allowing for efficient RSV infection.

Discussion

Insects are known to be vectors of many plant viruses, and some of these plant viruses rely entirely on insect vectors for their plant-to-plant transmission. Previous studies have suggested that insect salivary proteins may facilitate arbovirus infection of plants through various mechanisms, such as breaking down plant cell walls, inhibiting plant immune defenses and callose deposition, directly binding to arbovirus to prevent their detection by plant immune systems, and more. Further research is needed to fully understand the role that insect salivary proteins play in arboviruses infection of plants. In this study, we have found that RSV infection of rice plants triggers the deposition of callose. However, this process is reversed by a tripartite interaction involving the central component of an insect salivary protein LssaCA, which subsequently binds to RSV in insect salivary glands and then OsTLP in the plants. The activity of OsTLP's β -1,3-glucanase is significantly increased when this tripartite interaction occurs; leading to the degradation of callose and thus allowing for efficient RSV infection (Fig. 8).

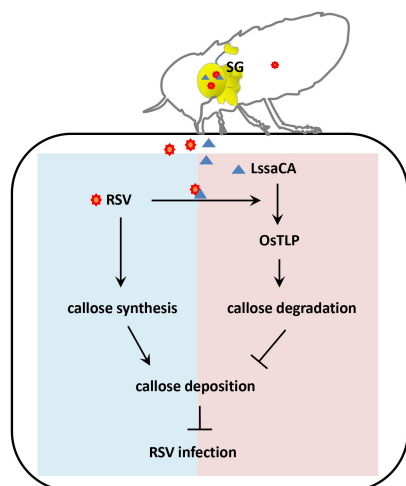


Figure 8.

Proposed model of how RSV NP-LssaCA-OsTLP tripartite interaction leads to callose degradation to promote RSV infection of plants.

A. L. striatellus feeding on rice plants inoculates both RSV and LssaCA into the phloem sieve elements. Within the plants, RSV induces callose synthesis, causing callose deposition to inhibit RSV infection. However, LssaCA reverses this inhibition by binding to OsTLP to enhance its β -1,3-glucanase activity, which degrades the deposited callose. The interaction between RSV and LssaCA further enhances the OsTLP enzymatic activity and facilitates RSV infection.

Callose is a complex polysaccharide, and its deposition in plant phloem plays an important role in defending against insect attacks and viral spread (Zavaliev et al., 2011). When attacked by piercing insects or viruses, plants activate callose deposition on the sieve plates to occlude the flow of phloem sap, thus discouraging feeding and reducing viral long-distance spread. This accumulation of callose in plant phloem can be switched on or off through the action of callose synthases and hydrolases (Li et al., 2012; Shangguan et al., 2018)). Studies have shown that expression levels of these glucanase enzymes differ between resistant and susceptible plant species. This suggests that some plants may be more successful at resisting infection due to high levels of glucanase enzymes in their tissues, which allow them to more quickly and effectively form callose barriers against insects and viruses (Hao et al., 2008). Some insects have evolved β -1,3-glucanases enzymes that can degrade the deposited callose, thus enabling them to bypass the plant's defense mechanisms and continue feeding (Bucher et al., 2001; Hao et al., 2008). In this study, we demonstrate a novel mechanism by which an insect saliva protein binds to a plant β -1,3-glucanases protein, thus activating the enzymatic activity and promoting callose degradation.

In this study, we found that plants fed on by viruliferous insects showed higher callose deposition than did plants fed on by uninfected insects (Fig. 5A–C), suggesting that arbovirus infection *via* insect saliva may represent a double stress to the plant, inducing stronger callose deposition that must be overcome for successful insect feeding and viral spread. To

analyze how this happens, we looked at the tripartite interaction between LssaCA-NP-OsTLP. We found that this interaction induced significantly higher β -1,3-glucanase activity than the LssaCA-OsTLP interaction alone, suggesting that SBPH and RSV have evolved mechanisms to increase β -1,3-glucanase activity in order to bypass the plant's defense of callose deposition and facilitate successful viral spread.

LssaCA is carbonic anhydrase, but its role in aiding RSV infection does not rely on its enzymatic activity (Fig. S2). Our results found that even though LssaCA mutants had significantly reduced enzymatic activity, their ability to facilitate RSV infection was not significantly decreased. Because LssaCA interacts with RSV NP, we measured the interactions between LssaCA mutants and NP. Pull-down assays indicated that all the mutants still interacted with RSV NP (Fig. S4), while the mutants with a stronger interaction (LssaCA^{H111D} and LssaCA^{E153H}) showed better facilitation for RSV infection levels (Fig. S2). These results further supported the importance of LssaCA-NP interaction in mediating RSV infection, while the exact role of LssaCA enzymatic activities in aiding RSV infection is still unclear.

In conclusion, this study elucidated a tripartite interaction among a plant virus, an insect saliva protein, and a host phloem β -1,3-glucanase. This interaction inhibited the plant defense of callose deposition and enabled arboviral phloem transmission. This study provides new insights into the tripartite virus-insect vector-plant interaction, which is relevant to many agriculturally important plant arboviruses whose transmission is facilitated *via* insect saliva proteins.

Figure S1. Nucleotide sequence and deduced amino acid sequence of LssaCA.

Underlining indicates the predicted signal peptide sequence. Seven predicted catalytically active residues are shown in red. Bold font indicates eukaryotic-type carbonic anhydrase sequences.

Figure S2. LssaCA enzymatic activity does not affect RSV infection.

A. Summary of the predicted enzymatic active sites and the muteins.

B. Relative esterase activities of recombinant LssaCA protein and its muteins. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.0001$.

C. Schematic diagram to show the process of microinjecting RSV into a plant. The RSV was pre-incubated with recombinantly expressed LssaCA, the LssaCA muteins, or a BSA control, before being microinjected into the rice phloem.

D. The effects of LssaCA, its muteins, or the BSA control, on the levels of RSV infection as determined by RT-PCR. The titers of RSV (copy number of NP) were measured at 14 dpi.

Figure S3. Nucleotide sequence and deduced amino acid sequence of OsTLP.

Underlining indicates the predicted signal peptide sequence. Bold font indicates the conserved thaumatin family protein sequence.

Figure S4. The molecular interaction between RSV and LssaCA muteins is important for RSV to infect rice plants.

A. Pull-down assays showing the interactions between the LssaCA muteins (expressed with a His-tag) and RSV NP (GST-NP). GST was used as a negative control. Anti-His and Anti-GST represent antibodies against 6*His and GST, respectively. Both LssaCA^{H111D} and LssaCA^{E153H} (marked in red) have a stronger interaction with NP compared to the wild-type LssaCA protein.

B. Preincubation of RSV with either LssaCA^{H111D} or LssaCA^{E153H} facilitates RSV infection of plants after being delivered via microinjection into the phloem. The titers of RSV (copy number of NP) were measured at 14 dpi. BSA was used as a negative control. *, $p < 0.05$; **, $p < 0.01$.

Table S1. Primers used in this study.

Materials and methods

Viruses, insects, host plants, and antibodies

The RSV-free and RSV-infected *L. striatellus* individuals used in this study were originally captured in Jiangsu Province, China, and were maintained in our laboratory. All plants used to rear *L. striatellus* were grown in a growth incubator (14.5 cm height, 3.2 cm radius) at 25°C under a 16-h light/8-h dark photoperiod. Insects were transferred to fresh seedlings every 15 days to ensure adequate nutrition and insect activity.

The RSV-specific antibody was produced using RSV ribonucleoproteins (RNPs) as antigens. The LssaCA polyclonal antibody was produced using recombinantly expressed LssaCA protein (with a His-tag at the C-terminus) as the antigen.

Tissue collection

Insect dissection and tissue collection were performed using previously described procedures (Huo et al., 2018). In brief, insects were anesthetized at 4°C for 10 min, and the forelegs were severed at the coxa-trochanter joint using forceps. The hemolymph was expelled and drawn to the tip of clean forceps. The insects were then dissected from the abdomen in pre-chilled PBS buffer. The fat body was collected with forceps, aided by the tension of the liquid. Tissues, including the guts, salivary glands, and ovaries, were washed three times in PBS to remove contaminants from the hemolymph.

Bioinformatic analysis

The signal peptide and transmembrane helix were predicted using tools at the SignalP 5.0 server (<https://services.healthtech.dtu.dk/service.php?SignalP-5.0>) and the TMHMM-2.0 server (<https://services.healthtech.dtu.dk/service.php?TMHMM-2.0>). Homology searches of the protein sequence of LssaCA and OstLP were conducted using the BLAST algorithm at the National Center for Biotechnology Information (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). Conserved domains were predicted with SMART (<http://smart.embl-heidelberg.de/>).

Collection of watery and gel saliva

Saliva was collected as described previously (Huo et al., 2022). Watery saliva was obtained by centrifuging the collected artificial diet at 10,000 ×g for 10 min to remove the solid gel. To collect gel saliva proteins, the membrane was washed twice with PBS. Sheathes adhering to the parafilm were denatured and solubilized in a solution consisting of 8 M urea, 4% v/v CHAPS, 0.1% w/v sodium dodecyl sulfate (SDS), and 2% v/v DTT. The parafilm was incubated on an orbital shaker at room temperature for 45 min. Then, the denatured gel saliva was concentrated to a final volume of 100 µL. Saliva samples were collected from 2,000 insects for silver staining and western blotting analyses.

Measurement of carbonic anhydrase activity

Carbonic anhydrase activity was characterized by measuring its esterase activity. *p*-nitroethyl acetate (*p*-NPA) was used as the substrate. The LssaCA enzyme solution was prepared as follows: LssaCA was expressed in *E. coli* with a His-tag at the C-terminus and the purified LssaCA protein was diluted in PBS buffer (pH 7.0) to a final concentration of 1 mg/mL. Details of the reaction conditions have been described elsewhere (Verpoorte et al., 1967). Briefly, 1 mL of freshly prepared 3 mM *p*-NPA dissolved in acetone was added to 1.9 mL 100 mM PBS (pH 7.0), followed by the addition of 0.1 mL LssaCA enzyme solution (0.5 mg/mL in 100 mM PBS). As the blank control, 2 mL 100 mM PBS (pH 7.0) was added to 1 mL 3 mM *p*-NPA. The absorbance at 384 nm was measured at the start of the reaction and 3 min later. One unit (U) of enzyme activity is defined as the amount of enzyme that hydrolyzes 1 μ mol *p*-nitroethyl acetate per minute at pH 7.0 at 25 °C.

Immunofluorescence assays

Immunofluorescence assays (IFA) were performed to determine the localization of RSV and LssaCA in *L. striatellus* salivary glands. The dissected salivary gland samples were fixed in 4% v/v paraformaldehyde at 4°C overnight and then permeabilized in 2% v/v Triton X-100 for 2 h. The samples were incubated with rabbit anti-RSV serum and mouse anti-LssaCA serum (1:250 dilution in PBST/FBS: PBS containing 2% v/v Triton X-100 and 2% fetal bovine serum) for 1 h at room temperature. After washing twice with PBST, the samples were incubated with 1:200 diluted Alexa Fluor 488-conjugated goat anti-mouse antibody and Alexa Fluor 594-conjugated goat anti-rabbit antibody (Invitrogen) for 1 h. After washing three times with PBST, the samples were examined under a Leica TCS SP8 microscope (Leica, Wetzlar, Germany).

Evaluation of callose deposition in rice plants

Five insects were allowed to feed on a 10 d-old rice plant in a leaf area of 2 × 2 cm for 24 h. The leaf samples were immersed in 5 mL alcoholic lactophenol (1 volume phenol: glycerol: lactic acid: water (1:1:1:1) and 2 volumes ethanol), then evacuated with a syringe for 15 min, and then incubated at 65°C for 30 min until the chlorophyll was completely cleared. Cleared leaf samples were rinsed with 50% v/v ethanol, then with water, and then stained in 150 mM K₂HPO₄ (pH 9.5) containing 0.01% w/v aniline blue for 30 min in the dark. Stained leaf samples were sectioned and mounted in 50% v/v glycerol on slides. The slides were examined under a Zeiss observer Z1 immunofluorescence microscope (Carl Zeiss MicroImaging GmbH, Gottingen, Germany), and fluorescence intensity was determined using ImageJ software.

Callose concentrations were measured using a plant callose ELISA kit (Catalog number: JL46407; Jonln, Shanghai, China), according to the manufacturer's protocol. The callose solution was prepared as follows: leaf material from the feeding site was homogenized with PBS buffer (pH 7.4) and the mixture was centrifuged at 500–1000 g for 20 min. The supernatant was collected for the detection of callose concentration. The absorbance at 450 nm was measured and callose concentrations were calculated based on the standard curve.

RNA interference

We conducted RNAi analyses to determine the function of LssaCA in mediating RSV infection and regulating callose deposition. The *LssaCA*-specific gene fragment was PCR-amplified with the primer pair LssaCA-T7F/LssaCA-T7R (Table S1). Then, dsRNA was synthesized using

a commercial kit (T7 RiboMAX Express RNAi System, Promega, Madison, WI, USA) and purified by phenol: chloroform extraction and isopropanol precipitation. Finally, 36.8 nL dsRNA at 1 ng/nL was delivered into the insect hemocoel for gene silencing. Microinjection was performed using a glass needle through a Nanoliter 2000 (World Precision Instruments, Sarasota, FL, USA). The negative control, GFP dsRNA, was synthesized and microinjected following the same protocol.

RT-qPCR

Gene transcript levels were determined by RT-qPCR. The RNA was extracted from *L. striatellus* tissues using an RNeasy Mini Kit (Qiagen, Hilden, Germany), and from plant leaves using TRIzol. Reverse transcriptional PCR (iScript cDNA Synthesis Kit, Bio-Rad, Hercules, CA, USA) and SYBR-Green-based qPCR (SYBR Green Supermix, Bio-Rad) was performed according to the manufacturer's protocols. The primer pair used to amplify *LssaCA* was *LssaCA*-q-F/*LssaCA*-q-R (Table S1). Viral RNA copies were measured using the primers NP-q-F/NP-q-R (Table S1). *L. striatellus* translation elongation factor 2 (*EF2*) and rice *actin* were amplified as internal controls to ensure equal loading of cDNA isolated from different samples. The primer pairs *EF2*-q-F/*EF2*-q-R and *actin*-q-F/*actin*-q-R (Table S1) were used to amplify *EF2* and *actin*, respectively. Water was used as a negative control.

Yeast two-hybrid assays

Yeast two-hybrid screening was performed to identify rice proteins interacting with *LssaCA*. A high-complexity rice cDNA library was fused to *Gal4 AD* and transformed into the yeast strain Y187. The *LssaCA* gene lacking the signal-peptide sequence (*LssaCA*^{SP-}) was fused to *Gal4 DNA-BD* and transformed into the yeast strain Y2HGold. The two yeast strains were co-cultured overnight to allow mating to create diploids. After library screening, positive clones were selected on quadruple dropout medium: SD/-Ade/-His/-Leu/-Trp supplemented with X-a-Gal and aureobasidin A (QDO/X/A), and prey plasmids were isolated for sequencing. To further confirm the interaction between *LssaCA* and *OsTLP*, *LssaCA*^{SP-} was cloned into the prey vector pDB-Leu, and the *OsTLP* gene lacking the signal-peptide sequence (*OsTLP*^{SP-}) was cloned into the bait vector pPC86. The two plasmids were co-transformed into the yeast strain Mav203 and the transformants were selected on SD/-His-Leu-Trp + 20 mM 3-AT medium.

Pull-down assays

Pull-down assays were performed to determine the molecular interactions between *LssaCA* and RSV NP, *LssaCA* mutants and RSV NP, *LssaCA* and *OsTLP*, and *OsTLP* and the *LssaCA*-NP complex.

The interaction between *LssaCA* and RSV NP was determined using GST pull-down analyses as described previously (Huo et al., 2014). Briefly, *LssaCA*^{SP-} was cloned into the pGEX-3x vector and *LssaCA* was expressed as a GST-fusion protein (GST-*LssaCA*). The NP gene was cloned into the pET-30a (+) expression vector and the protein was expressed with an N-terminal His-tag (NP-His). The GST-*LssaCA* construct was expressed in, and purified from, *E. coli* strain BL21. The GST-*LssaCA* protein or the GST control protein was bound to Glutathione Sepharose beads (Cytiva, Marlborough, MA, USA), then NP-His protein was added to the beads. The bead-bound proteins were separated by SDS-PAGE and detected by western blotting with antibodies against GST or His-tag.

We conducted His pull-down assays to determine the interactions between the *LssaCA* mutants and RSV NP. The gene sequences encoding *LssaCA* mutants (without the signal-peptide sequence) were each cloned into the pET-30a (+) expression vector, and each mutant

was expressed with an N-terminal His-tag. The *NP* gene was cloned into the pGEX-3x vector and NP was expressed as a GST-fusion protein (GST-NP). Each of the His-tag-fused LssaCA mutants was bound to Ni Sepharose (GE Healthcare, Uppsala, Sweden), then the GST-NP protein or the GST control protein was added to the beads. The bead-bound proteins were separated by SDS-PAGE and detected by western blotting with antibodies against GST or His-tag.

Next, MBP pull-down analyses were performed to determine the interaction between OsTLP and LssaCA. The *OsTLP^{SP}* sequence was cloned into pMAL-p2X to produce the MBP-fusion protein (MBP-OsTLP). The MBP-OsTLP protein or the MBP control protein was bound to amylose resin (New England Biolabs, Beverly, MA, USA), and then the LssaCA-His protein was added to the beads. To determine the interaction between OsTLP and the LssaCA-NP complex, the LssaCA-His and GST-NP proteins were preincubated before being added to the beads. The bead-bound proteins were separated by SDS-PAGE and detected by western blotting with the MBP antibody (Cell Signaling Technology, Danvers, MA, USA), His-tag antibody (Tiangen, Beijing, China) and GST antibody (Tiangen).

Microscale thermophoresis (MST) assays

We conducted MST assays to detect the interactions between LssaCA and NP, and between LssaCA and OsTLP. First, 10 μ M purified LssaCA^{SP} protein (His-fusion protein) was labelled with a Monolith NT Protein Labeling Kit RED-NHS (Nano Temper Technologies GmbH, München, Germany) using red fluorescent dye NT-647 N-hydroxysuccinimide (amine-reactive), according to the manufacturer's instructions. The binding assays were performed on a Monolith NT.115 Microscale Thermophoresis instrument (Nano Temper Technologies GmbH) using capillaries treated using the standard method. The labeled protein LssaCA^{SP} was added to serially diluted NP (GST-NP, GST was used as a negative control) or OsTLP^{SP} (MBP-fusion protein, MBP was used as a negative control). The initial concentration of each protein was 40 μ M with 0.1% (v/v) Tween 20. The KD Fit function of NanoTemper Analysis software version 1.2.214.1 was used for curve fitting and to calculate the dissociation constant (Kd).

RSV inoculation through midrib microinjection

Midrib microinjection was performed as described previously (Zhao et al., 2017). Briefly, 200 RSV-infected *L. striatellus* were homogenized with 200 μ L PBS (pH 7.0), and the homogenate was centrifuged twice at 8000 g for 10 min at 4°C. The crude virus extract was transferred to a new centrifuge tube. The concentration of RSV in the extract was measured using a Nanodrop One instrument. The RSV extract was adjusted to 1 mg/mL for microinjection. Rice seedlings were grown in an incubator for 2 weeks until the leaf veins became clearly visible. Then, 23 nL RSV extract was microinjected into the midrib. The RSV titers in plants were detected at 7–14 days post microinjection.

β -1,3-glucanase activity

β -1,3-glucanase hydrolyzes laminarin and cuts the β -1,3-glucoside bond to produce a reducing terminus. Therefore, the rate of reducing sugars production was used to calculate enzyme activity. One unit of β -1,3-glucanase activity was defined as the amount of enzyme that hydrolyses laminarin to generate 1 μ g reducing sugars per minute.

The β -1,3-glucanase activity of OsTLP was measured using a beta-1,3-glucanase microplate assay kit (Cat. ASK1028, Bioworld, Bloomington, MN, USA) according to the manufacturer's instructions.

To determine the enzymatic activity of the recombinant OsTLP proteins, the MBP-fusion OsTLP^{SP} protein or the MBP control protein was adjusted to 0.5 mg/mL. To determine the enzymatic activity of the *in vivo*-expressed OsTLP^{SP}, 0.1 g rice leaf material was homogenized with 1 mL assay buffer on ice, and then the mixture was centrifuged at 12,000g at 4°C for 10 min. The supernatant was transferred into a new centrifuge tube and kept on ice until detection of enzymatic activity.

For the assay, a 50-μL aliquot of the protein solution was mixed with laminarin substrate and incubated at 37°C for 30 min. The reaction was terminated by incubating the reaction tube in a boiling water bath for 10 min. The reaction supernatant was transferred into a microplate and incubated at 90°C for 10 min before measuring the absorbance at 540 nm.

Generation of transgenic plants

The full-length *OsTLP* gene was PCR-amplified with the primer pair TLP-F/TLP-R (Table S1), and cloned into the plant expression vector pCAMBIA-1300-eGFP to produce the plasmid pCAMBIA-1300-OsTLP. The plasmid pCAMBIA-1300-OsTLP was transformed into seedlings of the *Japonica* rice cultivar Nipponbare through *Agrobacterium*-mediated transformation as described previously (Hiei et al., 1994). Thirty independent T₀ transformants were obtained and were screened by germinating seeds from T₂ lines on medium containing 50 μg/ml hygromycin. The hygromycin-resistant homozygous seeds were used for further experiments. Seeds from homozygous lines transformed with the empty vector pCAMBIA-1300-eGFP were used as the negative control.

Statistical analyses

Graphs were constructed and statistical analyses were performed using Prism 9.0 software (GraphPad Software, San Diego, CA, USA). Data are expressed as the mean ± standard deviation (SD) calculated from three independent experiments. The significance of differences between two groups was determined by unpaired Student's t-test. Differences among three or more groups were compared using one-way analysis of variance (ANOVA).

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Author contributions

Conceptualization, Z.L.L., H.Y., Z.J.; Methodology, H.Y., Z.J. and Y.J.; Investigation, Z.J., Y.J., M.X.Y., H.Y.; Writing-Original Draft, H.Y. and Z.J.; Writing-Review & Editing, Z.L.L. and F.R.X.; Funding Acquisition, Z.L.L., H.Y. and F.R.X.; Supervision, Z.L.L., and F.R.X.

Declarations of competing interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

In this study, the authors identify an insect salivary protein participating viral initiate infection in plant host. They found a salivary LssCA promoting RSV infection by interacting with OsTLP that could degrade callose in plants. Furthermore, RSV NP bond to LssCA in salivary glands to form a complex, which then bond to OsTLP to promote degradation of callose.

The story focus on tripartite virus-insect vector-plant interaction, and is interesting. However, the study is too simple and poor-conducted. The conclusion is also overstated due to unsolid findings.

Major comments:

1. The key problem is that how long the LssCA functioned for in rice plant. Author declared that LssCA had no effect on viral initial infection, but on infection after viral inoculation. It is unreasonable to conclude that LssCA promoted viral infection based on the data that insect inoculated plant just for 2 days, but viral titer could be increased at 14 day post-feeding. How could saliva proteins, which reached phloem 12-14 days before, induce enough TLP to degrade callose to promote virus infection? It was unbelievable.

2. Lines 110-116 and Fig. 1, the results of viruliferous insect feeding and microinjection with purified virus could not conclude the saliva factor necessary of RSV infection, because these two tests are not in parallel and comparable. Microinjection with salivary proteins combined with purified virus is comparable with microinjection with purified virus.

The second problem is how many days post viruliferous insect feeding and microinjection with purified virus did author detect viral titers? in Method section, authors declared that viral titers was detected at 7-14 days post microinjection. Please demonstrate the days exactly.

The last problem is that how author made sure that the viral titers in salivary glands of insects between two experiments was equal, causing different phenotype of rice plant. If not, different viral titers in salivary glands of insects between two experiments of course caused different phenotype of rice plant.

3. The callose deposition in phloem can be induced by insect feeding. In Fig. 5H, why was the callose deposition increased in the whole vascular bundle, but not phloem? Could the transgenic rice plant directional express protein in the phloem? In Fig.5, why was callose deposition detected at 24 h after insect feeding? In Fig. 6A, why was callose deposition decreased in the phloem, but not all the cells of the of TLP OE plant? Also in Fig.6A and B, expression of callose synthase genes was required.

Reviewer #2 (Public Review):

There is increasing evidence that viruses manipulate vectors and hosts to facilitate transmission. For arthropods, saliva plays an essential role for successful feeding on a host and consequently for arthropod-borne viruses that are transmitted during arthropod feeding on new hosts. This is so because saliva constitutes the interaction interface between arthropod and host and contains many enzymes and effectors that allow feeding on a compatible host by neutralizing host defenses. Therefore, it is not surprising that viruses change saliva composition or use saliva proteins to provoke altered vector-host interactions that are favorable for virus transmission. However, detailed mechanistic analyses are

scarce. Here, Zhao and coworkers study transmission of rice stripe virus (RSV) by the planthopper *Laodelphax striatellus*. RSV infects plants as well as the vector, accumulates in salivary glands and is injected together with saliva into a new host during vector feeding.

The authors present evidence that a saliva-contained enzyme - carbonic anhydrase (CA) - might facilitate virus infection of rice by interfering with callose deposition, a plant defense response. In vitro pull-down experiments, yeast two hybrid assay and binding affinity assays show convincingly interaction between CA and a plant thaumatin-like protein (TLP) that degrades callose. Similar experiments show that CA and TLP interact with the RSV nuclear capsid protein NT to form a complex. Formation of the CA-TLP complex increases TLP activity by roughly 30% and integration of NT increases TLP activity further. This correlates with lower callose content in RSV-infected plants and higher virus titer. Further, silencing CA in vectors decreases virus titers in infected plants. Interestingly, aphid CA was found to play a role in plant infection with two non-persistent non-circulative viruses, turnip mosaic virus and cucumber mosaic virus (Guo et al. 2023 doi.org/10.1073/pnas.2222040120), but the proposed mode of action is entirely different.

While this is an interesting work, there are, in my opinion, some weak points. The microinjection experiments result in much lower virus accumulation in rice than infection by vector inoculation, so their interpretation is difficult. Also, the effect of injected recombinant CA protein might fade over time because of degradation or dilution. The authors claim that enzymatic activity of CA is not required for its proviral activity. However, this is difficult to assess because all CA mutants used for the corresponding experiments possess residual activity. It remains also unclear whether viral infection deregulates CA expression in planthoppers and TLP expression in plants. However, increased CA and TLP levels could alone contribute to reduced callose deposition.